



Olist E-Commerce Data Analysis

Overview

This project performs an Exploratory Data Analysis (EDA) on the Olist Brazilian E-Commerce Dataset. The objective is to analyze customer behavior, sales performance, product categories, payment methods, delivery efficiency, and customer satisfaction to generate business insights that can support data-driven decision making.

Objectives

- Perform data cleaning and preprocessing.
 - Merge multiple datasets into a single analytical dataset.
 - Analyze customer purchasing patterns.
 - Study sales and revenue trends.
 - Evaluate product category performance.
 - Analyze payment preferences.
 - Measure delivery efficiency.
 - Examine customer review patterns.
 - Generate actionable business insights using visualizations.
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Dataset

The project uses the **Olist Brazilian E-Commerce Public Dataset**, which contains real-world e-commerce transactions.

Datasets used:

- Customers Dataset
 - Orders Dataset
 - Order Items Dataset
 - Products Dataset
 - Sellers Dataset
 - Payments Dataset
 - Reviews Dataset
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Technologies Used

- Python
- Google Colab / Jupyter Notebook
- Pandas
- NumPy

- Matplotlib
 - Seaborn
 - SciPy
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Data Preprocessing

The following preprocessing steps were performed:

- Imported multiple CSV files.
 - Checked dataset dimensions.
 - Handled missing values.
 - Converted timestamp columns into datetime format.
 - Filtered only delivered orders.
 - Merged datasets using common keys.
 - Removed incomplete records.
 - Created new features such as:
 - Delivery Time
 - Delivery Delay
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Exploratory Data Analysis

The notebook performs several analyses, including:

Sales Analysis

- Order trend over time
- Monthly sales trend
- Revenue analysis

Customer Analysis

- Total unique customers
- Customer distribution by state
- Customer purchasing behavior

Product Analysis

- Revenue by product category
- Highest selling categories
- Lowest performing categories

Payment Analysis

- Payment method distribution
- Payment value analysis

Delivery Analysis

- Average delivery time
- Delivery delay analysis
- Product categories with highest delays

Customer Review Analysis

- Average review scores
 - Lowest rated product categories
 - Seller review analysis
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Visualizations

The project includes several visualizations such as:

- Line Charts
- Bar Charts
- Pie Charts
- Revenue Comparisons
- Customer Distribution Graphs
- Delivery Analysis Charts
- Product Category Rankings

These visualizations help identify business trends and customer behavior.

Key Insights

- Customer purchases show clear sales trends over time.
 - Certain product categories contribute significantly more revenue.
 - Credit card is the most commonly used payment method.
 - Delivery time varies across product categories.
 - Delayed deliveries often correspond with lower customer review scores.
 - Some sellers consistently receive lower ratings and may require quality improvements.
 - Revenue is concentrated within a limited number of high-performing categories.
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Project Workflow

1. Import libraries
2. Load datasets
3. Clean data
4. Merge datasets
5. Perform feature engineering
6. Conduct exploratory data analysis

7. Create visualizations
 8. Generate business insights
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Business Applications

This analysis can help businesses:

- Improve inventory planning
 - Optimize delivery operations
 - Enhance customer satisfaction
 - Increase revenue through category analysis
 - Monitor seller performance
 - Understand customer purchasing behavior
 - Improve payment strategies
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Future Improvements

- Build interactive dashboards using Power BI or Tableau.
 - Develop machine learning models for sales prediction.
 - Perform customer segmentation.
 - Implement recommendation systems.
 - Forecast demand using time-series models.
 - Analyze customer lifetime value (CLV).
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Conclusion

This project demonstrates how exploratory data analysis can transform raw e-commerce data into meaningful business insights. Through data preprocessing, visualization, and statistical analysis, the project highlights customer behavior, operational efficiency, product performance, and revenue trends. These findings can assist businesses in making informed strategic decisions and improving overall customer experience.

Repository Structure

```
├─ Olist_Ecommerce_Analysis.ipynb
├─ datasets/
│   └─ customers.csv
│   └─ orders.csv
│   └─ order_items.csv
│   └─ products.csv
│   └─ sellers.csv
```

```
|   ├── payments.csv
|   ├── reviews.csv
|   ├── images/
|   ├── README.md
|   └── requirements.txt
```

Author

Deeshanshi Sahu

Computer Science Engineering Student

Python | Data Analytics | Machine Learning | Data Visualization